

Research Project Report: Parking Lot Detection AI System

Prepared by: Taj Al-ajaji

Supervised by: T. Raghad Al-shammari

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1. Introduction

This project focuses on designing and implementing a machine learning system capable of automatically classifying parking lot images into two distinct categories: empty and not empty. The core methodology relies on leveraging digital image data and applying various machine learning algorithms to evaluate space availability in real time.

The primary objective of this research is to compare multiple predictive models to identify the most effective architecture for image classification tasks. Furthermore, this study demonstrates how machine learning can be practically integrated into real-world applications, such as smart parking systems and advanced traffic management.

2. Dataset Description

The dataset utilized in this empirical study was sourced from Kaggle: **Parking Lot Detection Counter Dataset**. It consists of a comprehensive collection of parking space images divided into two structural classes:

- **Empty:** Representing vacant parking spaces available for use.
- **Not_Empty:** Representing occupied parking spaces containing vehicles.

3. Data Preprocessing

To optimize model training and ensure reliable performance evaluation, several critical preprocessing steps were systematically applied to the raw image data:

- **Grayscale Conversion:** All images were converted to grayscale to minimize computational complexity and allow the models to focus strictly on essential structural features.
- **Image Resizing:** The spatial dimensions of the images were resized to a fixed, uniform input size of **64×64** pixels.
- **Numerical Array Conversion:** The processed images were transformed into multi-dimensional numerical arrays compatible with model training pipelines.
- **Data Splitting:** The final dataset was split into an **80%** training set to fit the models, and a **20%** testing set to rigorously evaluate generalization performance on unseen data.

4. Models Implemented

A diverse selection of machine learning and deep learning models were implemented and contrasted to determine the optimal classification approach:

- **Logistic Regression:** Implemented as a baseline linear model for binary classification.
- **Random Forest Classifier:** An ensemble learning methodology based on decision trees, utilized to improve predictive accuracy and mitigate overfitting.
- **K-Nearest Neighbors (KNN):** A distance-based instance classifier that categorizes sample images according to their closest training examples.
- **Neural Network:** A deep learning model built using the **Keras** framework, engineered to extract and model highly complex, non-linear patterns within the image data.

5. Model Evaluation Metrics

To comprehensively measure the effectiveness of each model in classifying parking lot spaces, five standard performance evaluation metrics were employed:

1. **Accuracy:** Quantifies the overall proportion of correct predictions made by the model.
2. **Precision:** Measures the exactness of the model by showing how many predicted positive cases were actually correct.
3. **Recall:** Measures the sensitivity of the model by capturing how many actual positive cases were correctly identified.
4. **F1-Score:** Computes the harmonic mean of precision and recall, offering a balanced performance evaluation metric.
5. **Confusion Matrix:** Utilized as a visual tool to map correct and incorrect predictions across both classes. Dedicated confusion matrix plots were generated for each model to audit detailed performance behavior.

6. What I Learned

Through the lifecycle of this project, several key architectural and practical insights were gained regarding image classification pipelines:

- **Model Complexity vs. Performance:** More complex models are not always better. While the Deep Learning Neural Network demonstrated stable learning, the classic **Random Forest Classifier** achieved the best overall performance and highest accuracy for this specific task.
- **The Impact of Preprocessing:** Proper data cleaning and preparation before training made the most significant difference. Converting images to grayscale and resizing them to a uniform dimension (**64×64**) drastically reduced computational overhead while preserving crucial features necessary for exact classification.

7. Conclusion

This project successfully demonstrated the utility of machine learning and neural networks in automating image-based parking lot classification. Based on the experimental findings, the following conclusions can be drawn:

- Among all tested frameworks, the **Random Forest Classifier** achieved the superior overall performance and yielded the highest classification accuracy.
- The **Neural Network** displayed a highly stable learning trajectory and solid generalization capabilities during both the training and validation phases.
- Overall, every implemented model demonstrated robust capability on the dataset, underscoring the profound effectiveness of machine learning methodologies in handling modern image classification tasks.